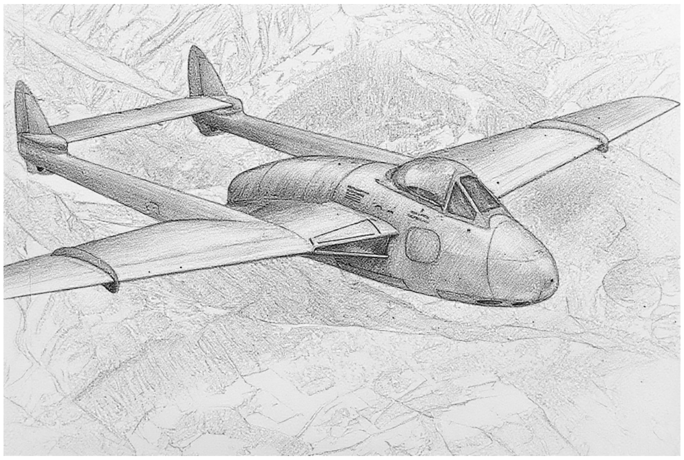
SNCASE Mistral



The SNCASE (Société nationale des constructions aéronautiques du Sud-Est) SE.535 Mistral was a day fighter and fighter-bomber. It was named for the strong wind that occurred frequently in Provence.

Versions

SE 535 Mistral

The SE 535 Mistral was a development of the Vampire FB.5 (previously license-built by SNCASE), but fitted with the more powerful Nene 104B motor (license-built by Hispano-Suiza) providing 5,000 lb of thrust compared to 3,000 lb for the Goblin 2 engine of the FB.5 and 3,500 lb for the Goblin 3 engine of the FB.6.

The Nene engine had been trialed in the Vampire F.2 with its need for greater airflow satisfied by additional "elephant ears" inlets on the upper side of the fuselage behind the cockpit. However, these intakes caused handling problems close to the critical Mach number. In the Vampire F.30, this was mitigated by moving the additional intakes to the underside of the fuselage. In contrast, the Mistral omitted them completely and instead opted for redesigned and enlarged wing-root inlets, and as a consequence had much more benign characteristics at high speed.

The Mistral also featured an ejection seat and air conditioning to facilitate its use in the French colonies in Africa.

It was not clear how much de Havilland was involved in the development of the Mistral, but nevertheless early prototypes of the Mistral were sometimes referred to as the Vampire FB.53.

The Mistral served in the French AA from 1953 until at least 1961.

Armament and Stores

Its gun armament was four 20 mm Hispano cannons with 150 rounds, as in the Vampire. Air-to-ground ordnance included rockets, HE bombs, and napalm bombs.

Combat

The Mistral saw combat with the French AA in the colonial wars in Tunisia and Algeria.

ADC

- Mistral

See Also

- de Havilland Vampire
- SNCASE Vampire

Photo Credit

- SNCASE Mistral: US Department of State (Public Domain)

Mistral

Crew: Pilot

Power APs/DPs/FPs: ○

	CL	1/2	DT	Fuel
AB	—	—	—	—
M	1.0	1.0	1.0	1.5
N	0.0	0.0	0.0	0.5
I	0.5	0.5	0.5	0.0
SPBR	0.5	0.5	0.5	—

Cruise Speed:	3.5	Restr. Arcs:	—
Climb Speed:	3.0	Blind Arcs:	30–
Visibility:	4	Internal Fuel:	200
Size:	+1	AtA Refuel:	No
Vulnerability:	+0	Ejection Seat:	Early

Maneuver HFPs/DPs:

LR/DR	1.0	1.0
VR		0.5

Turn DPs:

	CL	1/2	DT
TT	0.0	0.0	0.0
HT	1.0	1.0	1.0
BT	1.0	1.0	1.0
ET	—	—	—

Speeds and Ceilings / **Climb Capabilities**

Alt. Band	Conf. Ceil.	CL 48	1/2 44	DT 40	Dive Speed	CL AB	CL Oth	1/2 AB	1/2 Oth	DT AB	DT Oth	Alt. Band
EH+	46+	—	—	—	—	—	—	—	—	—	—	EH+
VH	36–45	2.5 – 5.5	3.0 – 5.0	3.0 – 4.5	6.0	—	0.5	—	0.5	—	0.5	VH
HI	26–35	2.0 – 5.5	2.5 – 5.0	2.5 – 4.5	6.5	—	0.5	—	0.5	—	0.5	HI
MH	17–25	1.5 – 5.5	2.0 – 5.0	2.0 – 5.0	6.5	—	0.5	—	0.5	—	0.5	MH
ML	8–16	1.5 – 5.5	1.5 – 5.0	1.5 – 5.0	6.5	—	1.0	—	1.0	—	1.0	ML
LO	0–7	1.0 – 5.5	1.5 – 5.0	1.5 – 5.0	6.0	—	1.5	—	1.0	—	1.0	LO

Radar:	—
ECCM:	—
Arcs:	—
Search:	—
Track:	—
Lock-On:	—

ECM:	
RWR:	—
DDS:	—
DJM:	—
AJM:	—
BJM:	—

Weapon Stations Diagram:

Guns:	Four 20 mm Hispano Mk V
To Hit:	7/4/3
Ammunition:	5.0
Gunsight:	TT+0/HT+1/BT+2
Ranging:	—
AtA/AtG:	5/6*

Bomb System: Manual (+0)

Technology:

None

Load Point Limits: CL : 0–2; 1/2: 3–4; DT : 5+

Weight Limit: 2,000

Station	Limit	Allowed Loads
1 and 6	1,000	BB FT
2–3 and 4–5	200	RK

Load Notes:

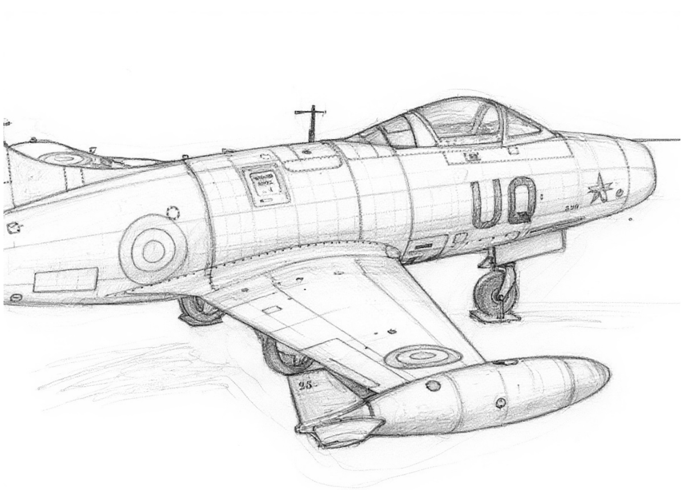
1. Stations 2 to 5 may each carry one or two RP-3 RKs.
2. May use 450L FTs.

Notes:

1. The SNCASE SE.535 Mistral is a day fighter and fighter bomber. It is derived from the SNCASE Vampire FB.5, but has a more powerful Nene engine, an ejection seat, and air conditioning for use in hot climates.
2. High transonic drag (HTD). Rapid acceleration (RA) if speed ≤ 4.0.

VPs: 6/4/2/1

Dassault Ouragan



The Dassault Ouragan was a fighter-bomber and the first French-designed jet-powered combat aircraft enter service. Like many of its contemporaries, it had a single engine, an air-intake in the nose, swept wings and tail, and wing-tip fuel tanks. Its name means “hurricane” in French.

Versions

MD 450A and MD 450B

The initial 450A version used a non-afterburning Rolls-Royce Nene 102 engine. The subsequent 450A version used a similar Rolls-Royce Nene 104 engine license-built by Hispano-Suiza and had a revised undercarriage. However, the MD 450B was otherwise almost identical to the MD 450A.

The Ouragan entered service with the Armée de l’Air in 1952, replacing Vampires and other older aircraft. Its service life was short, and in 1955 it was replaced by the Mystère IV.

The Ouragan also served with the Indian Air Force from 1953 and was known as the Toofani (“hurricane” in Hindi). It was withdrawn from front-line services in 1965.

It also served with the Israeli Air Force from 1953 and served until the 1970s. In 1975, several were sold on to El Salvador, where they served until the late 1980s.

Armament and Stores

The principal armament was four 20 mm Hispano-Suiza cannons under the nose. Two 450L fuel tanks would often be carried for extended range. For air-to-ground missions, the Ouragan could also carry bombs, napalm tanks, 105 mm Brand T-10 rockets, and SNEB 68 mm rocket pods.

Combat

The Ouragan saw combat with the Israeli Air Force in the Suez Crisis and the 1967 War. In Indian service, it saw combat in the 1962 China-India War and the 1965 India-Pakistan War.

ADC

- Ouragan

Photo Credit

- Ouragan: Ad Meskens

Ouragan

Power APs/DPs/FPs: ○

	CL	1/2	DT	Fuel
AB	—	—	—	—
M	1.0	1.0	1.0	1.0
N	0.0	0.0	0.0	0.5
I	0.5	0.5	0.5	0.0
SPBR	0.5	0.5	1.0	—

Cruise Speed:	4.0	Restr. Arcs:	—
Climb Speed:	3.0	Blind Arcs:	30–
Visibility:	5	Internal Fuel:	220
Size:	+0	AtA Refuel:	No
Vulnerability:	+1	Ejection Seat:	Early

Crew: Pilot

Maneuver HFPs/DPs:

LR/DR	1.0	1.0
VR		0.5

Turn DPs:

	CL	1/2	DT
TT	0.0	0.0	1.0
HT	1.0	1.0	2.0
BT	1.0	2.0	2.0
ET	—	—	—

Speeds and Ceilings						Climb Capabilities						
Alt. Band	Conf. Ceil.	CL 42	1/2 38	DT 34	Dive Speed	CL AB	CL Oth	1/2 AB	1/2 Oth	DT AB	DT Oth	Alt. Band
EH+	46+	—	—	—	—	—	—	—	—	—	—	EH+
VH	36–45	2.0 – 5.0	2.5 – 4.5	—	6.5	—	0.5	—	0.5	—	—	VH
HI	26–35	1.5 – 5.0	2.0 – 4.5	2.0 – 4.5	6.5	—	0.5	—	0.5	—	0.5	HI
MH	17–25	1.5 – 5.5	1.5 – 5.0	1.5 – 4.5	6.5	—	1.0	—	0.5	—	0.5	MH
ML	8–16	1.0 – 5.5	1.5 – 5.0	1.5 – 4.5	6.5	—	1.0	—	0.5	—	0.5	ML
LO	0–7	1.0 – 5.5	1.0 – 5.0	1.5 – 4.5	6.5	—	1.0	—	1.0	—	0.5	LO

Radar:	—
ECCM:	—
Arcs:	—
Search:	—
Track:	—
Lock-On:	—

ECM:	IFF
RWR:	—
DDS:	—
DJM:	—
AJM:	—
BJM:	—

Weapon Stations Diagram:

Guns:	Four 20 mm Hispano Mk V
To Hit:	7/4/3
Ammunition:	5.0
Gunsight:	TT+0/HT+1/BT+2
Ranging:	—
AtA/AtG:	5/6*

Bomb System: Manual (+0)

Technology:

None

Load Point Limits: CL : 0–2; 1/2: 3–6; DT : 7+

Weight Limit: 2,200

Station	Limit	Allowed Loads
1–4 and 7–10	150	RK
5 and 6	1,100	BB RP RK FT
11–12 and 13–14	200	RK

Load Notes:

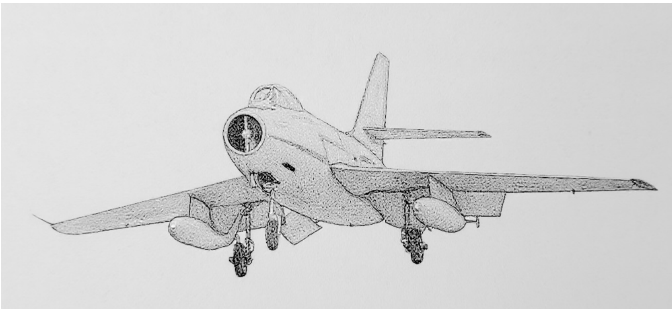
1. Either stations 1 to 4 and 7 to 10 or 11 to 14 may be used.
2. Stations 1 to 4 and 7 to 10 may each carry one or two T-10 RKs (large RK).
3. May use 450L FTs.

Notes:

1. The Dassault MD 450A/B Ouragan is a fighter-bomber.
2. High transonic drag (HTD).

VPs: 7/5/2/1

Dassault Mystère IV



The Mystère IV was a fighter-bomber. It resembled the earlier Mystère aircraft, but was a new design adapted for supersonic flight. It featured an air-intake in the nose, a single engine, and sharply swept wings and tail.

Versions

MD 454A

The MD 454A or Mystère IVA model was the only version to reach production. It featured a non-afterburning Rolls-Royce Tay engine (supplied by Rolls-Royce themselves or built under license by Hispano-Suiza) and two 30 mm DEFA cannons under the nose. It could reach supersonic speeds only in a dive.

The Mystère IVA entered service with the Armée de l'aire in 1955 and served first as an interceptor and then as a fighter-bomber. It also served with the Israeli Air Force from 1956 to 1971 and with Indian Air Force from 1957 to 1973.

Armament and Stores

The principal armament of the Mystère IVA was two 30 mm DEFA cannons under the nose. Two 450L fuel tanks would often be carried for extended range. For air-to-ground missions, the Mystère IVA could also carry bombs, napalm tanks, 105 mm Brand T-10 rockets, and SNEB 68 mm rocket pods.

Combat

Israeli Mystère IVAs saw combat in the 1956 and 1967 Arab-Israeli Wars. French aircraft also fought in the 1956 Suez Crisis. Indian aircraft fought in the 1965 and 1971 India-Pakistan Wars.

ADC

- Mystère IVA

Photo Credit

- Mystère IV: Anidaat (CC BY-SA 4.0)

Mystère IVA

Crew: Pilot

Power APs/DPs/FPs: ○

	CL	1/2	DT	Fuel
AB	—	—	—	—
M	1.0	1.0	1.0	1.0
N	0.0	0.0	0.0	0.5
I	0.5	0.5	0.5	0.0
SPBR	0.5	0.5	1.0	—

Maneuver HFPs/DPs:

LR/DR	1.0	1.0
VR		0.5

Turn DPs:

	CL	1/2	DT
TT	0.0	1.0	1.0
HT	1.0	1.0	1.0
BT	2.0	3.0	3.0
ET	—	—	—

Cruise Speed:	5.0	Restr. Arcs:	—
Climb Speed:	3.5	Blind Arcs:	30–
Visibility:	5	Internal Fuel:	150
Size:	+0	AtA Refuel:	No
Vulnerability:	+1	Ejection Seat:	Early

Speeds and Ceilings						Climb Capabilities						
Alt. Band	Conf. Ceil.	CL 44	1/2 40	DT 36	Dive Speed	CL AB	CL Oth	1/2 AB	1/2 Oth	DT AB	DT Oth	Alt. Band
EH+	46+	—	—	—	—	—	—	—	—	—	—	EH+
VH	36–45	2.5 – 6.0	3.0 – 5.5	3.0 – 5.0	6.5	—	0.5	—	0.5	—	0.5	VH
HI	26–35	2.0 – 6.0	2.5 – 5.5	2.5 – 5.0	7.0	—	0.5	—	0.5	—	0.5	HI
MH	17–25	2.0 – 6.5	2.0 – 5.5	2.5 – 5.0	7.0	—	1.0	—	0.5	—	0.5	MH
ML	8–16	1.5 – 6.5	2.0 – 6.0	2.0 – 5.5	7.0	—	1.0	—	1.0	—	0.5	ML
LO	0–7	1.5 – 6.5	1.5 – 6.0	2.0 – 5.5	7.0	—	1.0	—	1.0	—	1.0	LO

Radar:	APG-30
ECCM:	—
Arcs:	—
Search:	—
Track:	—
Lock-On:	7

ECM:	IFF
RWR:	—
DDS:	—
DJM:	—
AJM:	—
BJM:	—

Weapon Stations Diagram:

Guns:	Two 30 mm DEFA 552
To Hit:	6/3/2
Ammunition:	3.5
Gunsight:	TT+0/HT+1/BT+2
Ranging:	RE
AtA/AtG:	6/6

Bomb System: Manual (+0)

Technology:

None

Load Point Limits: CL : 0–2; 1/2: 3–6; DT : 7+

Weight Limit: 4,000

Station	Limit	Allowed Loads
1 and 4	1,500	BB RP
2 and 3	1,100	BB RP FT
5–8 and 9–12	200	RK

Load Notes:

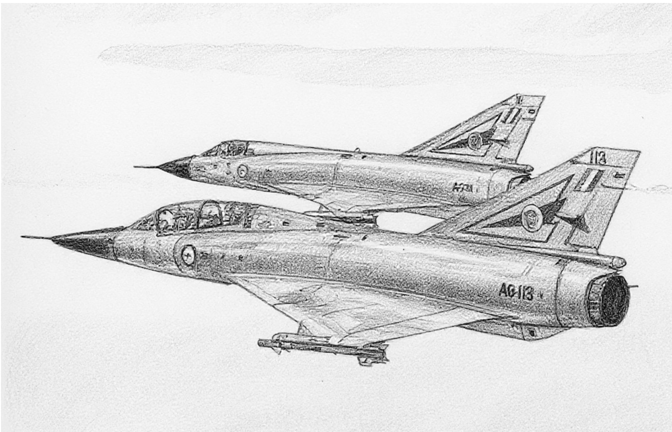
1. Either stations 1 to 4 or stations 5 to 12 may be used.
2. If stations 1 or 4 are loaded with more than 1,100 lb, the maximum turn rate is reduced to HT.
3. Stations 5 to 12 may each carry one or two 105 mm T-10 RKs (large RK).
4. May use 450L FTs.

Notes:

1. The Dassault MD 454 Mystère IVA is a fighter-bomber.
2. High transonic drag (HTD).
3. Hit rolls at low transonic speed or greater have a +1 modifier to the hit roll due to instability.

VPs: 10/7/3/2

Dassault Mirage III



The Dassault Mirage III is a supersonic interceptor and multirole aircraft. [1]

Versions

Mirage IIIE

The IIIE has an improved Cyrano-II radar, an extended fuselage to give room for more fuel and improved avionics including a RWR, and a slightly more powerful Atar 09C-3 engine.[2]

The IIIE first flew in 1961 and entered service with the French Armée de l'aire in 1964.[3]

Mirage IIIEA

The IIIEA is a version for the IIIE for the Argentinian FAA.[4]

The IIIEA saw combat in the South Atlantic War.[5]

Armament and Stores

The gun armament is two 30 mm DEFA cannons each with 125 rounds.[6]

For air-to-air missions, the central station can carry an R.530 RHM or IRM, and the two outer wing stations can carry R.550 Magic IRMs.

To increase endurance, 500L, 900L, 1100L, 1300L, or 1700L fuel tanks could be carried, although only the 500L and 900L tanks could be carried at supersonic speeds. Not all tanks were available to all users. The original French 500L tanks were fixed, but jettisonable versions were later developed by Israel.[7]

Combat

Argentine FAA Mirage IIIEAs saw combat in the 1982 South Atlantic War.

ADCs

- Mirage IIIEA

See Also

- Dassault Mirage 5

Notes

1. **Sources.**
   - Goebel, Air Vectors (Wayback)
   - Kettle, “South Atlantic War”, Clash of Arms Games, 2nd Edition, 2002.
   - Reid, “South African Air Force Mirage III and Cheetah: External fuel tanks,” October 2023 (Revision 0)
   - Wikipedia on the Mirage III
   - Wikipedia on the DEFA cannon
   - Wikipedia on the R.530
   - Wikipedia on the R.550
   - Wikipedia on the Martel

   Photo Credit.
   - Curt Eddings (Public Domain)
2. **IIIEA ADC.** The ADC is based on those in *Air Power Journal* 20 and 44.

   RWR. Goebel states the E gained an RWR. I assume RWR-A.

   Station Limits. The station limits on the earlier ADCs need revision. The Mirage III can carry the TK-500 MRT, JL-100/JL-500 RPTs, of 1700L FTs on the inner-wing station and 1300L FTs on the centerline station (Kettle and Reid). In *Air Strike*, the TK-500 MRT has a weight of 1100 lb and can carry up to 2200 lb of bombs, for a total of 3300 lb, and the JL-100/JL-500 RPTs have a weight of 900/3000 lb. The 1300L FT from the IIIS weights 2540 lb and the 1700L FT from the IIIS in *Air Power Journal* 39 weighs 3340 lb.

   The ADC for the IIIE in *Air Power Journal* 20 has:
   - centerline: 1800 lb
   - inner wing: 3000 lb
   - outer wing: 550 lb

   The ADC for the IIIE in *Air Power Journal* 44 has:
   - centerline: 2600 lb
   - inner wing: 1850 lb
   - outer wing: 370 lb

   The ADC for the IIIS in *Air Power Journal* 39 has:
   - centerline: 2550 lb
   - inner wing: 260 lb (for Falcons)
   - inner wing: 1230 lb but can be overloaded to 2540 lb for 1300L FT (limited to HT) or 3340 lb for 1700L FT (limited to TT).
   - outer wing: 370 lb

   Kettle has these limits for the IIIEA (and also for the Dagger):
   - centerline: 1180 kg = 2600 lb
   - inner wing: 840 kg = 1850 lb, but capable of carrying 1300L FT (2540 lb)
   - outer wing: 168 kg = 370 lb

   From these, I synthesize for the IIIC/E
   - centerline: 2600 lb (able to carry the 1300L FT)
   - inner wing: 1230 lb but can be overloaded to 2540 lb for the 1300L FT or TK-500 MRT (limited to HT) or 3400 lb for the 1700L FT or JL-500 RPT (limited to TT).
   - outer wing: 370 lb
3. **IIIE Service History and Combat.** Wikipedia.
4. **IIIEA ADC.** The ADC is based on the IIIE with stores from Kettle. I assume it did not have the rocket motor.
5. **IIIEA Service History and Combat.** Wikipedia.
6. **Guns.** The internal guns are 30 mm DEFA 552A cannons with 125 rounds each (Wikipedia on Mirage III and DEFA). At 1,100–1,500 rounds per minute, this corresponds to 2.5–3.4 ammunition points. The ADCs give 3.0 ammunition points.
7. **FTs.** The ADC for the IIIE in *Air Power Journal* 44 lists:
   - 500L FT (1025 lb, 2.0/1.5 load points, and 43 fuel points).
   - 1300L FT (2450 lb, 4.0/2.5 load points, and 112 fuel points)
   - 1700L FT (3195 lb, 7.0/4.0 load points, and 146 fuel points) FTs. It comments that the 500L cannot be jettisoned and the 1300L and 1700L limit the speed to subsonic. When carrying the 1300L and 1700L FTs on the inner-wing stations, the turn rates are limited to HT and TT, respectively.

   The ADC for the IIIS in *Air Power Journal* 39 lists:
   - 625L FT (1230 lb, 3.0/2.0 load points, and 52 fuel points).
   - 1300L FT (2540 lb, 4.0/3.0 load points, and 108 fuel points). It can be carried on the inner-wing stations, but reduces the maximum turn rate to HT.
   - 1700L FT (3340 lb, 5.0/3.5 load points, and 142 fuel points). It can be carried on the inner-wing stations, but reduces the maximum turn rate to TT.

   Reid mentions the following French tanks for South African Mirage IIICZ. I assume they were also available for French IIEs:
   - 500L fixed/supersonic for stations 2 and 4.
   - 600L jettisonable/subsonic for stations 2, 3, and 4.
   - 1300L jettisonable/subsonic for stations 2, 3, and 4.
   - 1700L jettisonable/subsonic for stations 2 and 4.

Mirage IIIEA

Crew: Pilot

Power APs/DPs/FPs: ○

	CL	1/2	DT	Fuel
AB	2.5	2.0	1.5	5.0
M	1.5	1.0	1.0	2.0
N	0.0	0.0	0.0	1.0
I	0.5	0.5	1.0	0.0
SPBR	0.5	0.5	1.0	—

Cruise Speed:	6.0	Restr. Arcs:	—
Climb Speed:	4.5	Blind Arcs:	30–
Visibility:	6	Internal Fuel:	285
Size:	+0	AtA Refuel:	No
Vulnerability:	+0	Ejection Seat:	Std

Maneuver HFPs/DPs:

LR/DR	1.0	1.0
VR		0.0

Turn DPs:

	CL	1/2	DT
TT	1.0	2.0	2.0
HT	2.0	3.0	3.0
BT	3.0	4.0	5.0
ET	4.0	—	—

Speeds and Ceilings						Climb Capabilities						
Alt. Band	Conf. Ceil.	CL 54	1/2 48	DT 40	Dive Speed	CL AB	CL Oth	1/2 AB	1/2 Oth	DT AB	DT Oth	Alt. Band
EH+	46+	4.5 – 14.0	—	—	15.0	0.5	0.0	—	—	—	—	EH+
VH	36–45	3.5 – 13.0	4.0 – 10.5	4.5 – 8.5	14.0	1.0	0.5	1.0	0.0	0.5	0.0	VH
HI	26–35	3.0 – 11.0	3.5 – 10.0	4.0 – 8.0	13.0	2.0	1.0	1.0	0.5	1.0	0.5	HI
MH	17–25	2.5 – 10.0	3.0 – 9.0	3.5 – 7.5	12.0	2.0	1.0	1.0	0.5	1.0	0.5	MH
ML	8–16	2.0 – 9.0	2.5 – 8.0	2.5 – 6.5	11.0	4.0	2.0	3.0	1.0	2.0	1.0	ML
LO	0–7	2.0 – 8.5	2.0 – 7.5	2.5 – 6.0	10.0	4.0	2.0	3.0	1.0	2.0	1.0	LO

Radar:	Cyrano II
ECCM:	0
Arcs:	180+
Search:	80–15
Track:	50–12
Lock-On:	7

ECM:	IFF
RWR:	—
DDS:	—
DJM:	—
AJM:	—
BJM:	—

Weapon Stations Diagram:

Guns:	Two 30 mm DEFA
To Hit:	6/3/2
Ammunition:	3.0
Gunsight:	TT+0/HT+0/BT+2
Ranging:	RE
AtA/AtG:	6/6

Bomb System: Manual (+0)

Technology:

None

Load Point Limits: CL : 0–4; 1/2: 5–8; DT : 9+

Weight Limit: 8,800

Station	Limit	Allowed Loads
1 and 5	550	IRM
2 and 4	3,400	BB DR FT MRT
3	2,600	BB RHM FT DR

Load Notes:

1. Turns limited to HT/TT if more than 2600/1300 lb carried on either of stations 2 or 4.
2. May carry two BBs or DRs on station 3.
3. May carry Matra R.550 Magic IRMs on stations 1 and 5
4. May carry Matra R.530 RHMs on station 3.
5. May carry 1300L FTs on station 2, 3, and 4 and 1700L FTs on stations 2 and 4.
6. No supersonic flight while carrying FTs.

Notes:

1. The Dassault Mirage IIIEA is single-seat fighter, intercepter, and strike aircraft. The EA version was specifically built for the Argentine air force.
2. High bleed rate (HBR). Low transonic drag (LTD). Poor supersonic maneuverability (PSSM).

VPs: 22/15/7/4

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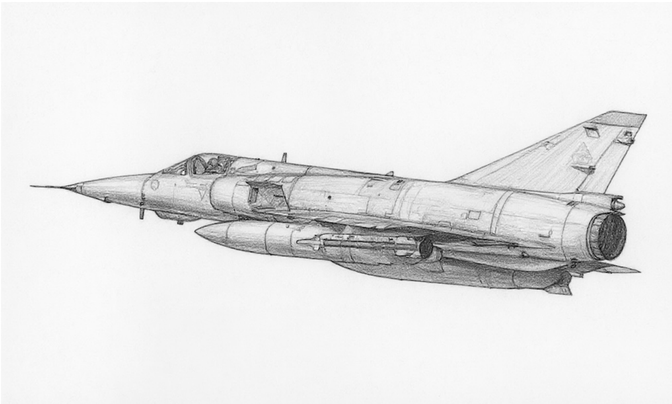
Dassault Mirage 5



The Dassault Mirage 5 (sometimes written as Mirage V) is a day fighter and strike aircraft. It first flew in 1967. Dassault developed it at the suggestion of the Israeli IAF, and to a large degree is a Mirage IIIE with the search radar and other all-weather avionics removed and with additional fuel and an additional pair of weapon stations. In some later versions, more compact electronics allowed all-weather capabilities to be restored.[1]

Versions

Mirage 5F

The Mirage 5F is the original version.[2]

These aircraft were originally built for Israel as the 5J. However, their delivery to Israel was blocked by the 1967 embargo, and they eventually entered service with the French Armée de l'air as the 5F and entered service around 1970.[3]

Mirage 5BA

The 5BA is a strike version for the Belgian Air Force. Changes included US avionics.[4]

The 5BA entered service in 1970, replacing F-84F Thunderstreaks, and was retired in 1993.[5]

Mirage 5BD

The 5BD is a two-seat trainer version for the Belgian Air Force, but it retains combat capability albeit without guns, like all two-seat versions of the Mirage 5.[6]

The 5BD entered service in 1970 and was retired in 1993.[7]

Mirage 5BR

The 5BR is a single-seat photoreconnaissance aircraft for the Belgian Air Force. It retains full combat capability.[8]

The 5BR entered service in 1970, replacing RF-84F Thunderstreaks, and was retired in 1993.[9]

Armament and Stores

The internal armament of the Mirage 5 is two 30 mm DEFA 552A cannons with 125 rounds each.[10]

The external weapon options are similar to the Mirage III, although RHMs are obviously not used on models without adequate radar.[11]

Combat

ADCs

- Mirage 5F
- Mirage 5BA
- Mirage 5BD
- Mirage 5BR

See Also

- Dassault Mirage III

Notes

1. **Sources.**
   - Goebel, Air Vectors (Wayback)
   - Mirage 5 Pilots Association, Home Page (Wayback)
   - Wikipedia on the Mirage 5
   - Wikipedia on the DEFA cannon
   - Wikipedia on the R.530
   - Wikipedia on the R.550
   - Wikipedia on the Martel
   - Wikipedia on the AS.30
   - Wikipedia on the Exocet
   - Wikipedia on the H-4 SOW
   - Wikipedia on the Haft-VIII Ra'ad
   - Wikipedia on the Haft-VIII Ra'ad-II

   Photo Credit.
   - Chris Lofting (GFDL 1.2)
2. **Mirage 5 ADF.** The original ADC for the Mirage 5, presumably the 5F version, appeared in *Air Strike*. Perleberg also has a version 2 ADC.

   The *Air Strike* errata sheet states that stations 3 and 5 can only carry BB.

   APJ 21 adds: ECCM 0; a standard ejection seat; LTD; PSSM; and allows the inner wing stations to be overloaded to 3000 lb.

   Perleberg's ADC allows the inner wing stations to be overloaded to 3300 lb with FT/RPT/MRT.

   The aircraft is limited to altitude 35 or less when DT, but the original *Air Strike* ADC has speed and climb data for the VH band (36–45) when DT. Perleberg removes this, and I follow him.

   The Mirage IIIE has the Cyrano-II search radar. When this was removed to create the Mirage 5, the simpler tracking/ranging AIDA-II radar used previously by the Étendard IV seems to have been installed.

   Jean-Baptiste Mouillet has commented that the 5F had no RWR and should have a manual bomb sight. However, the AIDA-II is noted as supplying both air-to-air and air-to-ground ranging, so I suspect a ballistic sight is more appropriate.
3. **Mirage 5F Service History.** The service history of the 5F is from Wikipedia.
4. **Mirage 5BA ADC.** The ADC for the 5BA is essentially that of the 5F, with some modifications to stores described below.
5. **Mirage 5BA Service History.** The service history of the 5BA is from the Mirage 5 Pilots Association and Wikipedia.
6. **Mirage 5BD ADC.** The ADC for the 5BD is essentially that of the 5BA, with the second crew added and the guns deleted. The Mirage IIIB lost its guns (Goebel), and I presume the 5BD did too.
7. **Mirage 5BD Service History.** The service history of the 5BD is from the Mirage 5 Pilots Association.
8. **Mirage 5BR ADC.** The ADC for the 5BR is essentially that of the 5BA with the addition of the camera nose. The Mirage IIIR retained its guns (Goebel), and I assume the 5BR did too. I assume the loss of the guns allows the aircraft to retain the same fuel capacity as the original 5BA.
9. **Mirage 5BR Service History.** The service history of the 5BR is from the Mirage 5 Pilots Association.
10. **Guns.** The internal guns are 30 mm DEFA 552A cannons with 125 rounds each (Wikipedia on Mirage V and DEFA). At 1,100–1,500 rounds per minute, this corresponds to 2.5–3.4 ammunition points. The ADCs give 3.0 ammunition points.
11. **Stores.** In the original *Air Strike* ADC, stations 1 to 5 are the same as on the Mirage III and stations 6 and 7 are under the rear fuselage. I have renumbered them conventionally from right to left.

    AAM. The *Air Strike* ADC states that it can use Matra R.530 RHMs on the centerline station, but I suspect this is only the air superiority variant with Agave radar. I suspect the 5F and others with the AIDA-II radar can only use IRMs. I have modified the ADC to reflect this.

    Belgian 5BA/5BD/5BR used Sidewinders (Mirage V Pilots Association). Belgium was not a user of the Matra R.530 or R.550 (Wikipedia on the R.530 and R.550).

    RPT/MTR. The JL100 RPT can carry 268L of fuel and 18 × 68 mm rockets. The TK500 MRT can carry 500L of fuel and 5 × 500 lb BB.

    BS. The original ADC of the 5F mentions BSes. Pakistani 5PA ROSE versions can carry the H-4 SOW BS. I can find no evidence that BSes were carried by others.

    RG. The original ADC of the 5F mentions RGs. I presume these refer to the AS.30/AS.30L RGs. The AS.30 was carried by Peruvian 5Ps (Wikipedia on AS.30), but can find no evidence that it was carried by others. I can also find no evidence that the AS.30L was carried by 5Fs or others.

    RS. The original ADC of the 5F mentions RSs. It is possible that Pakistani PAF Mirage 5Rs carry the Haft-VIII Ra'ad and Ra'ad-II cruise missile, which Pipes classifies as RS rather than ASM, but this did not exist when the original ADC was created. I do not know what this could refer to; perhaps the AJ-168 Martel?

    RK. The original ADC of the 5F mentions RKs. I can find no evidence that the 5 carried RKs.

    ASM. The original ADC of the 5F mentions ASMs. This may refer to the Pakistani 5PA3 which can carry the AM39 Exocet. I can find no evidence that other versions carried ASMs.

    ARM. The original ADC of the 5F mentions AS.37 ARMs. France was a user of the AS.37 Martel ARM (Wikipedia on the Martel), but I can find no evidence that the 5F carried it. Belgium was not a user of the AS.37 (Wikipedia on the Martel), so the 5BA/5BD/5BR most certainly did not.

Mirage 5F

Crew: Pilot

Power APs/DPs/FPs: ○

	CL	1/2	DT	Fuel
AB	2.5	2.0	1.5	5.0
M	1.5	1.0	1.0	2.0
N	0.0	0.0	0.0	1.0
I	0.5	0.5	1.0	0.0
SPBR	0.5	1.0	1.0	—

Cruise Speed:	6.0	Restr. Arcs:	—
Climb Speed:	4.5	Blind Arcs:	30–
Visibility:	6	Internal Fuel:	325
Size:	+0	AtA Refuel:	No
Vulnerability:	+0	Ejection Seat:	Std

Maneuver HFPs/DPs:

LR/DR	1.0	1.0
VR		0.0

Turn DPs:

	CL	1/2	DT
TT	2.0	2.0	2.0
HT	3.0	3.0	4.0
BT	4.0	5.0	5.0
ET	5.0	—	—

Speeds and Ceilings / **Climb Capabilities**

Alt. Band	Conf. Ceil.	CL 52	1/2 44	DT 35	Dive Speed	CL AB	CL Oth	1/2 AB	1/2 Oth	DT AB	DT Oth	Alt. Band
EH+	46+	4.5 – 13.5	—	—	14.0	0.5	0.0	—	—	—	—	EH+
VH	36–45	3.5 – 13.0	4.0 – 10.5	—	14.0	1.0	0.5	1.0	0.0	—	—	VH
HI	26–35	3.0 – 11.0	3.5 – 10.0	4.0 – 8.0	13.0	2.0	1.0	1.0	0.5	1.0	0.5	HI
MH	17–25	2.5 – 10.0	3.0 – 9.0	3.5 – 7.5	12.0	2.0	1.0	1.0	0.5	1.0	0.5	MH
ML	8–16	2.0 – 9.0	2.5 – 8.0	2.5 – 6.5	11.0	4.0	2.0	3.0	1.0	2.0	1.0	ML
LO	0–7	2.0 – 8.5	2.0 – 7.5	2.5 – 6.0	10.0	4.0	2.0	3.0	1.0	2.0	1.0	LO

Radar:	AIDA II
ECCM:	0
Arcs:	Limited
Search:	—
Track:	8–8
Lock-On:	7

ECM:	IFF
RWR:	B
DDS:	—
DJM:	—
AJM:	—
BJM:	—

Weapon Stations Diagram:

Guns:	Two 30 mm DEFA
To Hit:	6/3/2
Ammunition:	3.0
Gunsight:	TT+0/HT+0/BT+2
Ranging:	RE
AtA/AtG:	6/6

Bomb System: Ballistic (−1)

Technology:

None

Load Point Limits: CL : 0–6; 1/2: 7–9; DT : 10+

Weight Limit: 9,200

Station	Limit	Allowed Loads
1 and 7	550	BB BG RP RK IRM EP
2 and 4	3,400	BB BG RP RG RS RK GP DR FT RPT MRT
3 and 5	1,100	BB
4	2,600	BB BG RP RG RS RK GP PP EP WR WP FT IRM

Load Notes:

1. Turns limited to HT/TT if more than 2600/1300 lb carried on either of stations 2 or 6.
2. Stations 3 and 5 are under the rear fuselage.
3. May carry two BBs on station 4 without using a WR.
4. May carry AIM-9 and Matra 550 IRMs on stations 1 and 7
5. May carry Matra 530 IRMs on station 4.
6. May carry AS.30 RGs.
7. May carry TK-500 MRTs on stations 2 and 6.
8. May carry 500L, 600L, 1300L, or 1700L FTs on station 2 and 6 and 600L or 1300L FTs on station 4.
9. No supersonic flight while carrying 600L, 1300L, or 1700L FTs.

Notes:

1. The Dassault Mirage 5F is a single-seat day fighter-bomber aircraft originally built for the Israeli Air Force but which served in the French Armée de l'air.
2. High bleed rate (HBR). Low transonic drag (LTD). Poor supersonic maneuverability (PSSM).

VPs: 21/14/7/4

Mirage 5BA

Power APs/DPs/FPs: ○

	CL	1/2	DT	Fuel
AB	2.5	2.0	1.5	5.0
M	1.5	1.0	1.0	2.0
N	0.0	0.0	0.0	1.0
I	0.5	0.5	1.0	0.0
SPBR	0.5	1.0	1.0	—

Cruise Speed:	6.0	Restr. Arcs:	—
Climb Speed:	4.5	Blind Arcs:	30–
Visibility:	6	Internal Fuel:	325
Size:	+0	AtA Refuel:	No
Vulnerability:	+0	Ejection Seat:	Std

Crew: Pilot

Maneuver HFPs/DPs:

LR/DR	1.0	1.0
VR		0.0

Turn DPs:

	CL	1/2	DT
TT	2.0	2.0	2.0
HT	3.0	3.0	4.0
BT	4.0	5.0	5.0
ET	5.0	—	—

Speeds and Ceilings						Climb Capabilities						
Alt. Band	Conf. Ceil.	CL 52	1/2 44	DT 35	Dive Speed	CL AB	CL Oth	1/2 AB	1/2 Oth	DT AB	DT Oth	Alt. Band
EH+	46+	4.5 – 13.5	—	—	14.0	0.5	0.0	—	—	—	—	EH+
VH	36–45	3.5 – 13.0	4.0 – 10.5	—	14.0	1.0	0.5	1.0	0.0	—	—	VH
HI	26–35	3.0 – 11.0	3.5 – 10.0	4.0 – 8.0	13.0	2.0	1.0	1.0	0.5	1.0	0.5	HI
MH	17–25	2.5 – 10.0	3.0 – 9.0	3.5 – 7.5	12.0	2.0	1.0	1.0	0.5	1.0	0.5	MH
ML	8–16	2.0 – 9.0	2.5 – 8.0	2.5 – 6.5	11.0	4.0	2.0	3.0	1.0	2.0	1.0	ML
LO	0–7	2.0 – 8.5	2.0 – 7.5	2.5 – 6.0	10.0	4.0	2.0	3.0	1.0	2.0	1.0	LO

Radar:	AIDA II
ECCM:	0
Arcs:	Limited
Search:	—
Track:	8–8
Lock-On:	7

ECM:	IFF
RWR:	B
DDS:	—
DJM:	—
AJM:	—
BJM:	—

Weapon Stations Diagram:

Guns:	Two 30 mm DEFA
To Hit:	6/3/2
Ammunition:	3.0
Gunsight:	TT+0/HT+0/BT+2
Ranging:	RE
AtA/AtG:	6/6

Bomb System: Ballistic (–1)

Technology:

None

Load Point Limits: CL : 0–6; 1/2: 7–9; DT : 10+

Weight Limit: 9,200

Station	Limit	Allowed Loads
1 and 7	550	BB BG RP RK IRM EP
2 and 4	3,400	BB BG RP RK GP DR FT RPT MRT
3 and 5	1,100	BB
4	2,600	BB BG RP RK GP PP EP WR WP FT

Load Notes:

1. Turns limited to HT/TT if more than 2600/1300 lb carried on either of stations 2 or 4.
2. Stations 3 and 5 are under the rear fuselage.
3. May carry two BBs on station 4 without using a WR.
4. May carry AIM-9 IRMs on stations 1 and 7
5. May carry TK-500 MRTs on stations 2 and 6.
6. May carry 1300L or 1700L FTs on station 2 and 6 and a 1300L FT on station 4.
7. No supersonic flight while carrying 1300L or 1700L FTs.

Notes:

1. The Mirage 5BA is a single-seat day strike aircraft built for the Belgian Air Force.
2. High bleed rate (HBR). Low transonic drag (LTD). Poor supersonic maneuverability (PSSM).

VPs: 21/14/7/4

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Mirage 5BD

Crew: Pilot and Observer

Power APs/DPs/FPs: ○

	CL	1/2	DT	Fuel
AB	2.5	2.0	1.5	5.0
M	1.5	1.0	1.0	2.0
N	0.0	0.0	0.0	1.0
I	0.5	0.5	1.0	0.0
SPBR	0.5	1.0	1.0	—

Cruise Speed:	6.0	Restr. Arcs:	—
Climb Speed:	4.5	Blind Arcs:	30–
Visibility:	6	Internal Fuel:	325
Size:	+0	AtA Refuel:	No
Vulnerability:	+0	Ejection Seat:	Std

Maneuver HFPs/DPs:

LR/DR	1.0	1.0
VR		0.0

Turn DPs:

	CL	1/2	DT
TT	2.0	2.0	2.0
HT	3.0	3.0	4.0
BT	4.0	5.0	5.0
ET	5.0	—	—

Speeds and Ceilings						Climb Capabilities						
Alt. Band	Conf. Ceil.	CL 52	1/2 44	DT 35	Dive Speed	CL AB	CL Oth	1/2 AB	1/2 Oth	DT AB	DT Oth	Alt. Band
EH+	46+	4.5 – 13.5	—	—	14.0	0.5	0.0	—	—	—	—	EH+
VH	36–45	3.5 – 13.0	4.0 – 10.5	—	14.0	1.0	0.5	1.0	0.0	—	—	VH
HI	26–35	3.0 – 11.0	3.5 – 10.0	4.0 – 8.0	13.0	2.0	1.0	1.0	0.5	1.0	0.5	HI
MH	17–25	2.5 – 10.0	3.0 – 9.0	3.5 – 7.5	12.0	2.0	1.0	1.0	0.5	1.0	0.5	MH
ML	8–16	2.0 – 9.0	2.5 – 8.0	2.5 – 6.5	11.0	4.0	2.0	3.0	1.0	2.0	1.0	ML
LO	0–7	2.0 – 8.5	2.0 – 7.5	2.5 – 6.0	10.0	4.0	2.0	3.0	1.0	2.0	1.0	LO

Radar:	AIDA II
ECCM:	0
Arcs:	Limited
Search:	—
Track:	8–8
Lock-On:	7

ECM:	IFF
RWR:	B
DDS:	—
DJM:	—
AJM:	—
BJM:	—

Weapon Stations Diagram:

Guns:	—
To Hit:	—
Ammunition:	—
Gunsight:	TT+0/HT+0/BT+2
Ranging:	RE
AtA/AtG:	—

Bomb System: Ballistic (–1)

Technology:

None

Load Point Limits: CL : 0–6; 1/2: 7–9; DT : 10+

Weight Limit: 9,200

Station	Limit	Allowed Loads
1 and 7	550	BB BG RP RK IRM EP
2 and 4	3,400	BB BG RP RK GP DR FT RPT MRT
3 and 5	1,100	BB
4	2,600	BB BG RP RK GP PP EP WR WP FT

Load Notes:
1. Turns limited to HT/TT if more than 2600/1300 lb carried on either of stations 2 or 4.
2. Stations 3 and 5 are under the rear fuselage.
3. May carry two BBs on station 4 without using a WR.
4. May carry AIM-9 IRMs on stations 1 and 7
5. May carry TK-500 MRTs on stations 2 and 6.
6. May carry 1300L or 1700L FTs on station 2 and 6 and a 1300L FT on station 4.
7. No supersonic flight while carrying 1300L or 1700L FTs.

Notes:
1. The Mirage 5BA is a two-seat trainer and day strike aircraft built for the Belgian Air Force.
2. High bleed rate (HBR). Low transonic drag (LTD). Poor supersonic maneuverability (PSSM).

VPs: 21/14/7/4

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Mirage 5BR

Crew: Pilot

Power APs/DPs/FPs: ○

	CL	1/2	DT	Fuel
AB	2.5	2.0	1.5	5.0
M	1.5	1.0	1.0	2.0
N	0.0	0.0	0.0	1.0
I	0.5	0.5	1.0	0.0
SPBR	0.5	1.0	1.0	—

Cruise Speed:	6.0	Restr. Arcs:	—
Climb Speed:	4.5	Blind Arcs:	30–
Visibility:	6	Internal Fuel:	325
Size:	+0	AtA Refuel:	No
Vulnerability:	+0	Ejection Seat:	Std

Maneuver HFPs/DPs:

LR/DR	1.0	1.0
VR		0.0

Turn DPs:

	CL	1/2	DT
TT	2.0	2.0	2.0
HT	3.0	3.0	4.0
BT	4.0	5.0	5.0
ET	5.0	—	—

Speeds and Ceilings						**Climb Capabilities**						
Alt. Band	Conf. Ceil.	CL 52	1/2 44	DT 35	Dive Speed	CL AB	CL Oth	1/2 AB	1/2 Oth	DT AB	DT Oth	Alt. Band
EH+	46+	4.5 – 13.5	—	—	14.0	0.5	0.0	—	—	—	—	EH+
VH	36–45	3.5 – 13.0	4.0 – 10.5	—	14.0	1.0	0.5	1.0	0.0	—	—	VH
HI	26–35	3.0 – 11.0	3.5 – 10.0	4.0 – 8.0	13.0	2.0	1.0	1.0	0.5	1.0	0.5	HI
MH	17–25	2.5 – 10.0	3.0 – 9.0	3.5 – 7.5	12.0	2.0	1.0	1.0	0.5	1.0	0.5	MH
ML	8–16	2.0 – 9.0	2.5 – 8.0	2.5 – 6.5	11.0	4.0	2.0	3.0	1.0	2.0	1.0	ML
LO	0–7	2.0 – 8.5	2.0 – 7.5	2.5 – 6.0	10.0	4.0	2.0	3.0	1.0	2.0	1.0	LO

Radar:	AIDA II
ECCM:	0
Arcs:	Limited
Search:	—
Track:	8–8
Lock-On:	7

ECM:	IFF
RWR:	B
DDS:	—
DJM:	—
AJM:	—
BJM:	—

Weapon Stations Diagram:

Guns:	Two 30 mm DEFA
To Hit:	6/3/2
Ammunition:	3.0
Gunsight:	TT+0/HT+0/BT+2
Ranging:	RE
AtA/AtG:	6/6

Bomb System: Ballistic (–1)

Technology:

None

Load Point Limits: CL : 0–6; 1/2: 7–9; DT : 10+

Weight Limit: 9,200

Station	Limit	Allowed Loads
1 and 7	550	BB BG RP RK IRM EP
2 and 4	3,400	BB BG RP RK GP DR FT RPT MRT
3 and 5	1,100	BB
4	2,600	BB BG RP RK GP PP EP WR WP FT

Load Notes:

1. Turns limited to HT/TT if more than 2600/1300 lb carried on either of stations 2 or 4.
2. Stations 3 and 5 are under the rear fuselage.
3. May carry two BBs on station 4 without using a WR.
4. May carry AIM-9 IRMs on stations 1 and 7
5. May carry TK-500 MRTs on stations 2 and 6.
6. May carry 1300L or 1700L FTs on station 2 and 6 and a 1300L FT on station 4.
7. No supersonic flight while carrying 1300L or 1700L FTs.

Notes:

1. The Mirage 5BR is a single-seat photoreconnaissance and day strike aircraft built for the Belgian Air Force.
2. High bleed rate (HBR). Low transonic drag (LTD). Poor supersonic maneuverability (PSSM).
3. Oblique and overhead cameras installed in the nose.

VPs: 21/14/7/4